

Apollo Diagnostics - Laboratory Investigation Report

123, Nungambakkam High Road, Chennai - 600034 | NABL Accredited | Tel: 044-2827-5000

Patient Name:	Lalitha Bose	Report ID:	APL-HYPO-45-LAL-2026
Age / Gender:	45 Yrs / Female	Category:	Hypothyroidism
Weight / Height:	75 kg / 163 cm	BMI:	28.2 kg/m2
Ref. Doctor:	Dr. Ravi Kumar	Report Date:	10-Feb-2026

1. COMPLETE BLOOD COUNT [Circulatory System]

Red Cell Indices

Test	Result	Unit	Reference Range	Flag
Haemoglobin (Hb)	10.4	g/dL	12.0 - 16.0	LOW
Packed Cell Volume (PCV)	39.0	%	36 - 46	Normal
MCV	90.0	fL	83 - 101	Normal
MCH	29.0	pg	27 - 32	Normal
RDW-CV	13.0	%	11.6 - 14.0	Normal

Haematology Comment: Haemoglobin 10.4 g/dL - normocytic anaemia. MCV and MCH within range. Consider anaemia of chronic disease, early iron deficiency or haemolysis.

White Cells & Platelets

Test	Result	Unit	Reference Range	Flag
Total WBC	7000	cells/cmm	4500 - 11000	Normal
Neutrophils	62	%	40 - 75	Normal
Lymphocytes	28	%	20 - 45	Normal
Platelet Count	2.8	lakh/cmm	1.5 - 4.0	Normal

Comment: WBC and platelet counts within reference limits. No significant leucocytosis, leucopenia or thrombocytopenia.

2. IRON STUDIES [Haematopoietic System]

Test	Result	Unit	Reference Range	Flag
Serum Iron	110	ug/dL	60 - 170	Normal
TIBC	310	ug/dL	250 - 370	Normal
Transferrin Saturation	35.0	%	20 - 50	Normal
Serum Ferritin	80.0	ng/mL	12 - 150	Normal

Iron Studies Comment: Iron stores and transport markers are within normal limits. No iron deficiency or iron overload.

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3. HEPATIC FUNCTION TEST [Hepatobiliary System]

Liver Enzymes

Test	Result	Unit	Reference Range	Flag
SGPT / ALT	22	U/L	7 - 56	Normal
SGOT / AST	20	U/L	10 - 40	Normal
Alkaline Phosphatase (ALP)	70	U/L	44 - 147	Normal
Total Bilirubin	0.6	mg/dL	0.2 - 1.2	Normal
Albumin	4.2	g/dL	3.5 - 5.0	Normal

Hepatologist Note: Liver enzymes within normal limits. No hepatocellular damage or cholestatic process detected.

4. RENAL FUNCTION TEST [Urinary System]

Test	Result	Unit	Reference Range	Flag
Serum Creatinine	1.1	mg/dL	0.6 - 1.1	Normal
Blood Urea Nitrogen	21	mg/dL	7 - 20	HIGH
eGFR (CKD-EPI)	98	mL/min/1.73m2	60 - 120	Normal
Serum Uric Acid	4.8	mg/dL	2.4 - 6.0	Normal
Sodium (Na+)	139	mEq/L	135 - 145	Normal
Potassium (K+)	4.0	mEq/L	3.5 - 5.1	Normal

Nephrologist Note: eGFR 98 mL/min/1.73m2 - glomerular filtration well preserved. Renal function normal.

5. LIPID PROFILE [Cardiovascular Risk]

Test	Result	Unit	Reference Range	Flag
Total Cholesterol	218	mg/dL	0 - 200	HIGH
HDL Cholesterol	48	mg/dL	50 - 999	LOW
LDL Cholesterol	190	mg/dL	0 - 100	HIGH
Triglycerides	238	mg/dL	0 - 150	HIGH

Cardiologist Note: LDL 190 mg/dL (target <100 mg/dL) and Total Cholesterol 218 mg/dL are elevated. Triglycerides 238 mg/dL. Increased cardiovascular risk. RECOMMENDATION: Mediterranean diet, aerobic exercise, statin therapy consideration if >10-year CV risk >7.5%.

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6. THYROID FUNCTION TEST [Endocrine System]

Test	Result	Unit	Reference Range	Flag
TSH	20.2	uIU/mL	0.4 - 4.0	HIGH
Free T3	2.2	pg/mL	2.3 - 4.2	LOW
Free T4	0.7	ng/dL	0.8 - 1.8	LOW

Endocrinologist Comment: TSH: 20.2 uIU/mL is elevated - consistent with hypothyroidism. FT4 0.7 ng/dL is low (overt hypothyroidism). Thyroid antibodies (Anti-TPO) and ultrasound recommended. Consider levothyroxine therapy.

7. DIABETES SCREENING [Endocrine - Pancreas]

Test	Result	Unit	Reference Range	Flag
Fasting Blood Glucose	88	mg/dL	70 - 100	Normal
Post-Prandial Glucose (2hr)	115	mg/dL	0 - 140	Normal
HbA1c	5.1	%	0 - 5.7	Normal
HOMA-IR	1.5		0 - 2.5	Normal

Endocrinology Note: Glucose metabolism normal. HbA1c 5.1% - no diabetes or pre-diabetes. HOMA-IR 1.5 - no insulin resistance.

8. VITAMINS & BONE [Nutritional / Skeletal]

Test	Result	Unit	Reference Range	Flag
Vitamin D (25-OH)	16.0	ng/mL	30 - 100	LOW
Vitamin B12	450	pg/mL	211 - 911	Normal
Folate (Serum)	12.0	ng/mL	5.4 - 999	Normal
Calcium (Total)	8.2	mg/dL	8.5 - 10.5	LOW
PTH	32	pg/mL	15 - 65	Normal

Nutritionist/Endocrinologist Comment: Vitamin D: 16 ng/mL - deficient (10-20). PTH at 32 pg/mL is within range. Supplementation: 60,000 IU cholecalciferol weekly x 8 weeks, then 1000-2000 IU/day. Recheck in 3 months.

9. INFLAMMATORY MARKERS [Immune System]

Test	Result	Unit	Reference Range	Flag
ESR	23	mm/hr	0 - 20	HIGH
CRP	3.9	mg/L	0 - 5	Normal
hs-CRP	0.6	mg/L	0 - 1.0	Normal

Immunology Comment: CRP 3.9 mg/L and ESR 23 mm/hr are elevated. hs-CRP 0.6 mg/L - low. Inflammatory markers elevated - correlate with clinical symptoms and autoimmune screen.

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CLINICAL SUMMARY & RECOMMENDATIONS

- [ABNORMAL]** Anaemia: Hb=10.4 g/dL (LOW). Normocytic - check iron stores and chronic disease.
- [ELEVATED]** LDL=190 mg/dL (HIGH). Total Cholesterol=218. Dyslipidaemia - dietary + pharmacological management.
- [ABNORMAL]** Hypothyroidism: TSH=20.2 uIU/mL. Levothyroxine therapy evaluation required.
- [DEFICIENT]** Vitamin D=16 ng/mL (LOW). Supplementation required.
- [ACTION]** Next appointment: Dr. Ravi Kumar in 4-8 weeks for follow-up and review of supplementation response.

Report reviewed by: Dr. Anitha Krishnamurthy, MD (Pathology) | Verified by Dr. Ravi Kumar

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